

Finite definable choice from one measurable cardinal

Prikry tail recoding, finite equivariant rules,
and a sharp local ultrafilter threshold

Prepared for Vladimir Reshetnikov

18 September 2026

Abstract. Let W satisfy ZFC, let U be a normal measure on κ in W , and let $V = W[c]$ be its Prikry extension. Write $q_* = [c]$ for the class of the generic cofinal sequence modulo finite symmetric difference. We prove that, even allowing q_* and *arbitrary ground-model set parameters*, there is no nonempty finite definable family of proper selectors on finite subsets of $Q_\kappa = D_\kappa / \equiv^*$. More generally, a finite permutation-equivariant labeling problem has a nonempty finite definable family of solutions if and only if each individual permutation fixes a label; when this condition holds, a single solution is definable from a low-rank parameter.

This answers the Prikry-extension part of the explicit third-round question in the supplied synthesis. The proof replaces ultraexacting embeddings by a same-universe comparison: deleting one sufficiently late generic point preserves the generic tail class and the entire extension, while rotating finite residue configurations. Deciding the *set of finite tables*, rather than a chosen member of a finite family, makes the argument work without an illicit definable choice.

We also prove that definable traces on q_* are clopen in its relative Prikry topology and obtain full cardinality in every integer phase, with full-sized coordinate projections. Finally, no nonempty finite definable family of ultrafilters on q_* exists, but a countably infinite such family does exist. The latter construction is a general ZFC fact about tail classes. It is not a construction of countably many global ultrafilter-valued kernels.

Main advance and limits

The finite-selector and finite-label *obstructions*, previously obtained in the archive from ultraexactingness, already occur in an ordinary Prikry extension of one measurable cardinal. Their proofs even tolerate the parameter q_* and every fixed ground-model parameter.

This is a result about a specific Prikry model, not a proof that abstract rigidity alone implies these obstructions. No inconsistency of an ordinary large-cardinal axiom, no new lower consistency bound, and no worldwide priority claim are asserted.

Detailed proofs in English. The accompanying finite tests check only finite permutation algebra; the new set-theoretic arguments are not Lean formalizations.

Contents

1	The result and its place in the research program	3
2	Prikry forcing and the same-universe recoding lemma	4
3	Finite cyclic configurations and finite families	6
4	The exact finite-label classification	8
5	Selectors, tournaments, and linear orders	10
6	Relative topology and full phase saturation	11
7	Rigidity and the one-measurable upper bound	13
8	A sharp countable threshold for local ultrafilters	14
9	Proof audit and the remaining boundary	17
A	Finite verification and reproducibility	18

1 The result and its place in the research program

1.1 The question that is answered

The supplied *Large Cardinals Synthesis* [1] distinguishes two sources of definability obstructions. Its rigidity arguments already have a one-measurable Prikry construction. Its finite-family selector theorem and the necessity direction of the finite-label classification were proved using ultraexacting embeddings. The last question of its Section 15 asks whether these latter conclusions also hold in a Prikry extension of a measurable. The answer proved here is affirmative.

The source question also describes this as asking whether the conclusions follow from rigidity alone. Those are different logical assertions. A theorem in every normal Prikry extension establishes the first, but does not establish the second. Our proofs use a concrete finite-tail recoding property, not just regularity in a relativized HOD.

The distinction matters for interpreting the large-cardinal frontier. Ultraexactingness has the consistency strength of I0 by Aguilera–Bagaria–Goldberg–Lücke [4]. The present theorem shows that these particular finite-choice consequences have a construction using only one measurable cardinal. It does not reduce the consistency strength of ultraexactingness itself.

1.2 Statement in quotient language

For an uncountable cardinal λ of cofinality ω , put

$$D_\lambda = \{a \subseteq \lambda : \text{ot}(a) = \omega, \sup a = \lambda\}, \quad Q_\lambda = D_\lambda / \equiv^*,$$

where $a \equiv^* b$ means that $a \Delta b$ is finite. We identify an element of D_λ with its increasing enumeration when writing $a(n)$. For a set X , the notation $X^{(m)}$ means the ordered m -tuples of pairwise distinct elements of X .

A parameter $q \in Q_\lambda$ means the equivalence class as *one set*, not a chosen representative and not permission to use all its members as parameters. Thus $\text{OD}_{p,q}$ denotes definability from p, q , and finitely many ordinals. For $\text{OD}_{V_\lambda, q}$, finitely many individual parameters in V_λ are allowed. All definability and cardinalities below are computed in the final universe V , unless another model is displayed.

Finite labeling data are coded by hereditarily finite sets: we take the finite label set F to be a finite subset of ω with the specified action. This convention is part of the low-rank definability assertions; renaming arbitrary labels would also require their names as parameters.

Theorem 1.1 (Main theorem). Let $W \models \text{ZFC}$, let $U \in W$ be a normal measure on κ , and let $V = W[c]$ be the normal Prikry extension. Let $q_* = [c] \in Q_\kappa$. Then the following assertions hold in V .

1. For every $p \in W$, every $n \geq 2$, and every $0 < r < n$, there is no nonempty finite OD_{p,q_*} family of functions

$$f : [Q_\kappa]^n \longrightarrow [Q_\kappa]^r, \quad f(B) \subseteq B.$$

2. Let $m \geq 1$, let $\Gamma \leq S_m$, and let F be a finite Γ -set. The following are equivalent:

- (a) for some $p \in W$, there is a nonempty finite OD_{p,q_*} family of Γ -equivariant maps $Q_\kappa^{(m)} \rightarrow F$;
- (b) there is one such map in OD_{V_κ} ;
- (c) $\text{Fix}_F(g) \neq \emptyset$ for every $g \in \Gamma$.

3. Every OD_{p,q_*} subset of q_* , for $p \in W$, is clopen in the relative Prikry topology. If it is nonempty, then it has κ members in every integer phase relative to any $a \in q_*$. Eventually, each coordinate projection of each phase has cardinality κ .

4. The least size of a nonempty $\text{OD}_{V_\kappa, q_*}$ family of ultrafilters on the *full* power set of q_* is exactly $\aleph_0$.

Moreover κ is a strong limit of cofinality ω in V , $V_\kappa^V = V_\kappa^W$, and q_* is rigid in the sense of the supplied synthesis.

Part (2) concerns quotient tuples. A function on tuples of representatives that is invariant under finite changes of each coordinate factors uniquely through these quotient tuples. Thus this is the same labeling problem as in the source reports. The extension here is both the Prikry-model proof of necessity and the allowance of a *finite family* of candidate rules.

Topic	Supplied synthesis	This continuation
Finite selectors	Ultraexacting proof; Prikry case explicitly open	No finite family in every normal Prikry extension, even relative to q_* and ground parameters
Finite labels	Single-rule classification at an ultraexacting cardinal	Same criterion in a Prikry extension, also for finite families
Full fibres	Embedding-based phase and coordinate analysis	Relative clopenness and full-sized phase-coordinate projections at q_*
Ultrafilter selection	Finite obstructions; countable global-kernel question	Exact countable threshold for <i>local families</i> , with the global question kept separate

2 Prikry forcing and the same-universe recoding lemma

2.1 Classical input and conventions

Here a normal measure is nonprincipal and κ -complete, with the usual normality property for regressive functions. A condition in $\mathbb{P}_U^W$ is (s, A) , where s is a finite increasing sequence of ordinals below κ , $A \in U$, and all members of A exceed the last member of s . The empty stem is permitted. We use the convention that $(t, B) \leq (s, A)$ is stronger: t end-extends s , its new entries belong to A , and $B \subseteq A$. For an increasing cofinal sequence d , write

$$G_d = \{(s, A) \in \mathbb{P}_U^W : s \text{ is an initial segment of } d, \text{ran}(d) \setminus \text{ran}(s) \subseteq A\}.$$

We call this the filter *realized by* d .

Input A (Normal Prikry forcing). Normal Prikry forcing preserves cardinals, changes $\text{cf}(\kappa)$ to ω , and adds no bounded subsets of κ . For an increasing cofinal $d : \omega \rightarrow \kappa$, the Mathias criterion says

$$G_d \text{ is generic over } W \iff \text{for every } A \in U, d(n) \in A \text{ for all sufficiently large } n.$$

The extension is $W[G_d] = W[d]$.

The criterion is due to Mathias [2]. A convenient source for the preservation statements and the criterion is Benhamou [3, Corollaries 3.12, 3.15; Theorem 3.17]; the numbering here refers to the cited arXiv version. These are the only specialized forcing facts imported in the new proofs. We also use the ordinary forcing theorem and truth lemma.

Since a measurable cardinal is strongly inaccessible in its ground model, these facts give $V_\kappa^W = V_\kappa^V$. Here is the small extra argument, which is sometimes obscured by the phrase “no bounded subsets.”

Induct on $\alpha < \kappa$. If $V_\alpha^W = V_\alpha^V$, a ground-model bijection of V_α^W with an ordinal below κ codes every subset of that rank by a bounded subset of κ . Hence no new subset of V_α appears. At limit ranks take unions. The required ground-model size bounds follow from inaccessibility. The same preservation facts show that κ remains a strong limit cardinal.

2.2 Finite changes preserve the whole extension

Lemma 2.1 (Finite-tail recoding). Suppose $d \in D_\kappa$ and $d =^* c$. Then d is Prikry generic over W ,

$$W[d] = W[c] = V, \quad [d] = [c] = q_*.$$

If $P = (s, A) \in G_c$ and d is obtained by deleting a point of c after its stem s , then $P \in G_d$ as well.

Proof. For every $A \in U$, the sequence c is eventually in A . A finite change preserves this assertion, so the Mathias criterion applies to d . The finite sets $c \setminus d$ and $d \setminus c$ consist of ordinals and therefore belong to the transitive ground model W . Consequently $c \in W[d]$ and $d \in W[c]$, giving equality of the two extensions. The quotient classes are equal by their definition in this common universe.

For the last assertion, deleting a point beyond the stem leaves s an initial segment and leaves every remaining point beyond s in A . This is exactly the condition $P \in G_d$. $\square$

Remark 2.2 (What the lemma does not say). There is no asserted nonidentity automorphism of the transitive universe V , and no elementary embedding of V is being constructed. We compare two generic filters for the same ground-model forcing that produce the *same* extension. A set defined in that extension from fixed ground parameters, ordinals, and q_* is literally the same set under both presentations. The generic sequence used to evaluate that set can change.

Lemma 2.3 (Deciding a finite observation). Fix ground parameters p and ordinals $\vec{\alpha}$. Suppose a formula uniquely defines an object Z in V from $p, q_*, \vec{\alpha}$. Suppose a finite observation $O(d, Z)$ is defined from the increasing sequence d and Z , and always belongs to a fixed finite ground-model set E in a suitable forcing neighborhood of G_c . Then there are $P = (s, A) \in G_c$ and $e \in E$ such that

$$O(d, Z) = e$$

for every $d =^* c$ with $P \in G_d$. Here the object Z is the same actual object in V throughout.

Proof. By the truth lemma, choose a condition in G_c forcing the stated uniqueness and the properties needed to form the observation. Replace q_* by the canonical name $[\dot{c}]$ in its definition. Strengthen inside G_c to a condition P deciding the finite-valued name for $O(\dot{c}, Z)$, say with value e .

If $d =^* c$ realizes P , Lemma 2.1 says that d is generic and that its extension and tail-class parameter are the original V, q_* . The forced uniqueness thus identifies the object defined using $[d]$ with the original Z . Since $P \in G_d$, the decided observation has the value e there too. $\square$

This lemma is the replacement for the embedding interface in the source reports. The observation will be a *set of finite tables*. In particular, we will not choose a definable member of a definable finite family. Such a choice is precisely what must not be assumed.

3 Finite cyclic configurations and finite families

3.1 Realizing an arbitrary finite permutation

Let Ω be a finite set and τ a permutation of Ω . Enumerate its cycles as

$$C_j = (u_{j,0}, \dots, u_{j,d_j-1}), \quad j < b, \quad \tau(u_{j,i}) = u_{j,i+1 \bmod d_j}.$$

For an increasing cofinal sequence a , define

$$A_{j,i}(a) = \{a(d_j k + i) + j : k < \omega\}, \quad x_{u_{j,i}}(a) = [A_{j,i}(a)] \in Q_\kappa. \quad (3.1)$$

The addition is ordinal addition by a finite number. It remains below the uncountable cardinal κ .

Lemma 3.1 (Residues with distinct offsets). For $a =^* c$, the classes $x_u(a)$, $u \in \Omega$, are pairwise distinct. If d is obtained from a by deleting one point, then

$$x_u(d) = x_{\tau u}(a) \quad (u \in \Omega). \quad (3.2)$$

Both statements are unaffected by any exceptional finite initial segment.

Proof. A normal measure contains the club of infinite limit ordinals, so all sufficiently late entries of a are limit ordinals. The finitely many translations by $j < b$ are then separated: if $\alpha < \beta$ are two such limit ordinals, $\alpha + j < \beta$ for every $j < b$. For a fixed entry of a , different offsets give different ordinals. Hence the sets assigned to different cycles are eventually disjoint. Within a cycle, different residue classes use disjoint entries of a . All the sets in (3.1) are cofinal of order type ω . Distinct sets in this construction therefore have infinite symmetric difference.

If the deleted entry has index h , then $d(n) = a(n+1)$ for all $n \geq h$. For a residue $i < d_j - 1$, this advances the residue to $i+1$. For the last residue, it advances to residue 0 in the next block. The discarded initial pieces are finite. Taking quotient classes gives (3.2). $\square$

The normal-measure assertion used here has an elementary verification. If a club C were not measure one, the map $\alpha \mapsto \sup(C \cap \alpha)$ would be regressive on the measure-one set of sufficiently large $\alpha \notin C$. Normality would make it constant on a measure-one set, which would then be bounded below the next point of C . This contradicts nonprincipality and κ -completeness.

3.2 A family is represented by a set of tables

Let $m \geq 1$ and let F be finite. Suppose $\mathcal{L}$ is a nonempty finite OD_{p,q^*} family of maps

$$L : Q_\kappa^{\langle m \rangle} \longrightarrow F, \quad p \in W.$$

For the configuration from (3.1), define

$$t_L^a : \Omega^{\langle m \rangle} \longrightarrow F, \quad t_L^a(u_0, \dots, u_{m-1}) = L(x_{u_0}(a), \dots, x_{u_{m-1}}(a)),$$

and let

$$T(a) = \{t_L^a : L \in \mathcal{L}\}.$$

Distinct maps can have the same finite table. This causes no problem: $1 \leq |T(a)| \leq |\mathcal{L}|$.

Lemma 3.2 (Finite-table symmetry). Let $M = |\mathcal{L}|$ and let Ω, τ be any finite configuration with $|\Omega| \geq m$. The set $T(c)$ of tables is invariant under the permutation

$$R_\tau(t)(u_0, \dots, u_{m-1}) = t(\tau u_0, \dots, \tau u_{m-1}).$$

Consequently, for $H = \text{lcm}(1, 2, \dots, M)$,

$$t(\tau^H u_0, \dots, \tau^H u_{m-1}) = t(u_0, \dots, u_{m-1}) \quad \text{for every } t \in T(c). \quad (3.3)$$

Proof. The set of all maps $\Omega^{\langle m \rangle} \rightarrow F$ is finite. Its power set is also a finite ground-model set. Apply Lemma 2.3 to the observation $T(a)$, with $Z = \mathcal{L}$. Obtain a condition $P \in G_c$ deciding $T(\dot{c})$ to be the actual set $T(c)$.

Delete one entry of c after the stem of P , obtaining d . Then d realizes P , so $T(d) = T(c)$. On the other hand, the family $\mathcal{L}$ is unchanged, while Lemma 3.1 gives

$$t_L^d = R_\tau(t_L^c) \quad (L \in \mathcal{L}).$$

Thus $T(c) = R_\tau[T(c)]$. The map R_τ is injective on all tables, so its restriction to the nonempty finite set $T(c)$ is a permutation.

Every cycle of this permutation has length at most $|T(c)| \leq M$ and hence a length dividing H . Therefore R_τ^H fixes each individual table. The explicit formula for this assertion is (3.3). $\square$

Remark 3.3 (Why the exponent is indispensable). A finite definable family can be permuted without any one of its members being fixed. Already two binary choices on a two-point set are exchanged by the transposition. It is invalid to deduce a contradiction merely from a nontrivial cycle on the inputs. The exponent H kills the permutation of the finite set of tables; we then arrange that τ^H still acts nontrivially on a suitable input. This is the finite-family amplification principle of the source archive, now supplied with a forcing interface.

3.3 Amplification for any prescribed permutation

Use the standard left action of $\Gamma \leq S_m$ on ordered tuples:

$$(g \cdot \vec{z})_i = z_{g^{-1}(i)}.$$

A rule is equivariant when $L(g \cdot \vec{z}) = g \cdot L(\vec{z})$.

Theorem 3.4 (Elementwise fixed labels are necessary, even for finite families). If a nonempty finite OD_{p,q_*} family of equivariant maps $Q_\kappa^{\langle m \rangle} \rightarrow F$ exists, for $p \in W$, then every $g \in \Gamma$ has a fixed point in F .

Proof. Fix $g \in \Gamma$. Write its cycles on $\{0, \dots, m-1\}$ as

$$(i_{j,0}, \dots, i_{j,\ell_j-1}), \quad j < b, \quad g(i_{j,s}) = i_{j,s+1 \bmod \ell_j}.$$

Let M be the size of the family and $H = \text{lcm}(1, \dots, M)$. Construct a finite set Ω with cycles

$$C_j = \{(j, t) : t \in \mathbb{Z}/(H\ell_j)\mathbb{Z}\}, \quad \tau(j, t) = (j, t+1).$$

Define an injection $v : m \rightarrow \Omega$ by

$$v(i_{j,s}) = (j, Hs).$$

It has the crucial property

$$\tau^H v(i) = v(g(i)). \quad (3.4)$$

Realize Ω, τ using Lemma 3.1, and use Lemma 3.2 for the given family. Choose any one table $t \in T(c)$; this is an ordinary existential choice in the proof, not a definable selection from the family. Let

$$y = t(v(0), \dots, v(m-1)) \in F.$$

The table inherits coordinate equivariance from its underlying rule. By (3.3) and (3.4),

$$\begin{aligned} y &= t(\tau^H v(0), \dots, \tau^H v(m-1)) \\ &= t(v(g(0)), \dots, v(g(m-1))) \\ &= g^{-1} \cdot t(v(0), \dots, v(m-1)) = g^{-1} \cdot y. \end{aligned}$$

The inverse in the third line comes from our displayed convention for the left action on tuples. This equality is equivalent to $g \cdot y = y$. Since g was arbitrary, the conclusion follows. $\square$

The forcing decision is made only after the finite amplified configuration has been selected. Thus its stem length, however large, cannot spoil the argument: one deletes a point still later. No uniform bound on deciding conditions is needed.

4 The exact finite-label classification

The sufficiency argument is present in the supplied synthesis [1, Section 12.5]. We include its full proof to make the new Prikry classification self-contained and to delimit the new contribution. Its use of cyclic stabilizers also explains why “every element has a fixed point” cannot be replaced by “the whole group has a common fixed point.”

4.1 Incidence words and their stabilizers

Let $\vec{a} = (a_0, \dots, a_{m-1})$ be representatives of pairwise different classes in Q_λ . The finite union $\bigcup_i a_i$ has order type ω : below each ordinal smaller than λ , each a_i has only finitely many points. Enumerate the union increasingly as $\langle \delta_n : n < \omega \rangle$, and form the word

$$w_{\vec{a}}(n) = \{i < m : \delta_n \in a_i\} \in \mathcal{P}(m) \setminus \{\emptyset\}.$$

Each label occurs infinitely often, and each pair of distinct labels is separated infinitely often. The latter assertion expresses the fact that $a_i \Delta a_j$ is infinite.

Call words w, z *shift-tail equivalent*, written $w \sim_{\text{st}} z$, if there exist $r, s < \omega$ such that $w(r+n) = z(s+n)$ for every $n < \omega$. Finite changes in the representatives change the incidence word only by this equivalence: past the largest edited ordinal, the unions and their incidence labels agree. Permuting coordinates commutes with taking the incidence word.

Lemma 4.1 (Cyclic stabilizers of word classes). Let w be a word with the two infinite-occurrence and separation properties just stated. The stabilizer of $[w]_{\text{st}}$ under any finite $\Gamma \leq S_m$ is cyclic.

Proof. Consider the subgroup

$$B_w = \{(d, g) \in \mathbb{Z} \times \Gamma : w(n+d) = g \cdot w(n) \text{ for all sufficiently large } n\}.$$

The requirement “sufficiently large” ensures that negative shifts are well-defined. Composition and inversion of eventual identities show explicitly that B_w is a subgroup of the direct product.

Its projection to $\mathbb{Z}$ has trivial kernel. Indeed, if $(0, g) \in B_w$, then eventually $w(n) = g \cdot w(n)$. If g moved i , the labels i and $g^{-1}(i)$ would have the same membership in every sufficiently late $w(n)$, contradicting infinite separation. Hence g is the identity.

The projection identifies B_w with a subgroup of $\mathbb{Z}$. Such a subgroup is cyclic, including the possible trivial subgroup. Its image under the second projection is exactly the stabilizer of $[w]_{\text{st}}$: an eventual identity with an integer shift is equivalent to a shift-tail equivalence. An image of a cyclic group is cyclic. $\square$

4.2 Constructing a rule when the test passes

Proposition 4.2 (Sufficiency in ZFC). Let λ be any uncountable cardinal of cofinality ω . Let $\Gamma \leq S_m$ act on a finite set F , and suppose every $g \in \Gamma$ fixes some element of F . Then an equivariant rule $Q_\lambda^{(m)} \rightarrow F$ is definable from a well-order of the reals and finite codes for Γ, F . These parameters belong to V_λ .

Proof. Words over the finite alphabet $\mathcal{P}(m) \setminus \{\emptyset\}$ are coded by reals in a fixed way. Fix a well-order $<$ of the reals and a finite ordering of F . All these objects have rank below $\omega + \omega$, hence below the uncountable cardinal λ .

For each Γ -orbit of shift-tail classes of valid words, choose the class z_0 containing the $<$ -least real code in the union of those classes. This is a definable choice from the real codes, not a choice of cofinal representatives. By Lemma 4.1, the stabilizer $H \leq \Gamma$ of z_0 is cyclic. A generator h fixes a label by the hypothesis. Every power of h fixes that same label, so F^H is nonempty. Let y_0 be its first member in the fixed finite ordering.

For a class $z = g \cdot z_0$, define $\ell(z) = g \cdot y_0$. This is well-defined. If also $z = g' \cdot z_0$, then $g'^{-1}g \in H$, and hence $g \cdot y_0 = g' \cdot y_0$. The formula is visibly equivariant.

Finally, for $\vec{q} \in Q_\lambda^{(m)}$, take any representatives $\vec{a}$ and set

$$L(\vec{q}) = \ell([w_{\vec{a}}]_{\text{st}}).$$

The preceding discussion shows independence of every representative and coordinate equivariance. The construction gives a unique value by a formula using only the stated parameters, so Separation and Replacement produce the required set-function. Choice of representatives was used only to verify that the formula has a value; no representative enters the definition. $\square$

Theorem 4.3 (Prikry classification, including finite families). In the Prikry setup, the three assertions in Theorem 1.1(2) are equivalent.

Proof. The implication from a finite family to the fixed-point condition is Theorem 3.4. The condition yields a single OD_{V_κ} rule by Proposition 4.2. Its low-rank parameters are all in W because $V_\kappa^V = V_\kappa^W$. The singleton containing this rule is therefore a finite OD_{p, q_*} family for a suitable combined ground parameter p . $\square$

The phrase “for some $p \in W$ ” is important in the sufficiency statement. For a *prescribed* parameter p , no well-order of the reals is assumed to be definable from p, q_* . Necessity holds for every prescribed ground parameter. Sufficiency may adjoin the one explicitly stated low-rank well-order parameter.

Remark 4.4 (The five-label example). Let $\Gamma = S_3$, and let F be the disjoint union of the natural three-point action and the two-point sign action. The identity fixes every label. A transposition fixes the remaining point of the natural action. A three-cycle acts trivially on the sign action, so fixes both of its labels. Thus every element has a fixed label. No label is fixed by the entire group: the three-point action has no common fixed point, and a transposition exchanges the two sign labels. The classification nevertheless supplies a rule. This example, already identified in the archive, remains a useful check against an incorrectly strengthened necessity argument.

5 Selectors, tournaments, and linear orders

For $0 < r < n$, set

$$\text{Sel}_{n,r}(Q) = \{f : [Q]^n \rightarrow [Q]^r : (\forall B \in [Q]^n) f(B) \subseteq B\}.$$

These sets are nonempty in the ambient ZFC universe: any well-order of Q provides such selectors. Our obstruction concerns definable *families* of them, not their ordinary existence.

Theorem 5.1 (No finite definable family of proper selectors). For every ground parameter $p \in W$, every $n \geq 2$, and every $0 < r < n$, there is no nonempty finite family $\mathcal{F} \subseteq \text{Sel}_{n,r}(Q_\kappa)$ in OD_{p,q_*} .

Proof. Each selector f canonically determines a rule on ordered distinct tuples:

$$L_f(x_0, \dots, x_{n-1}) = \{i < n : x_i \in f(\{x_0, \dots, x_{n-1}\})\} \in [n]^r. \quad (5.1)$$

With $\Gamma = S_n$ acting naturally on $F = [n]^r$, this rule is equivariant. Indeed, reindexing the same unordered input by a permutation changes the indices of the selected points by exactly that permutation.

If $\mathcal{F}$ were a nonempty finite OD_{p,q_*} family, its image under the explicit map $f \mapsto L_f$ would be a nonempty finite definable family of equivariant rules. Theorem 3.4 would say that every permutation of n fixes an r -element subset of n . But an n -cycle is transitive: any nonempty invariant subset contains the entire orbit of any one of its elements, and hence is all of n . No proper nonempty r -element subset is invariant. This contradicts $0 < r < n$. $\square$

5.1 A direct view of the amplification

For this particular theorem one can see the last part of the proof without ordered labels. If $|\mathcal{F}| = M$, put $H = \text{lcm}(1, \dots, M)$ and take a single cycle of nH quotient classes. Encode each selector by its table on all n -subsets of these classes. After deciding the finite set of tables, deleting one late generic point rotates that set of tables. Its H th power fixes every table. On the block

$$B_0 = \{0, H, 2H, \dots, (n-1)H\},$$

the input rotation by H is an n -cycle. A fixed table must therefore select a subset of B_0 invariant under this transitive cycle, so it selects either none or all of B_0 . This is incompatible with a proper r -selection. The larger configuration is what prevents a finite family from evading the argument by permuting its members.

Corollary 5.2 (Order-theoretic consequences). There is no nonempty finite OD_{p,q_*} family of tournaments on Q_κ , of linear orders on Q_κ , or of choice functions on its nonempty finite subsets. This holds for every $p \in W$.

Proof. A tournament canonically selects the winner of each two-element subset. A linear order canonically selects its smaller element. A finite-set choice function restricts to a selector on two-element subsets. Each of these is a definable map from the relevant class of objects to $\text{Sel}_{2,1}(Q_\kappa)$. The image of a nonempty finite definable family under such a map is again a nonempty finite definable family, contradicting Theorem 5.1. It is immaterial if several original objects have the same image. $\square$

In particular these obstructions hold in OD_{V_κ} and, more strongly, in $\text{OD}_{V_\kappa, q_*}$. The additional parameter q_* does not supply a privileged representative or a starting index for its tail.

6 Relative topology and full phase saturation

The same recoding lemma has a local consequence that is stronger than cardinality alone. It identifies an actual neighborhood contained in every definable trace through each of its points.

6.1 Clopen definable traces

For $P = (s, A) \in \mathbb{P}_U^W$, define the basic subset of q_*

$$\mathcal{N}(s, A) = \{d \in q_* : (s, A) \in G_d\}.$$

These sets form a basis: every $d \in q_*$ realizes conditions, and two conditions realized by the same d have a common stronger condition realized by d . In detail, use the longer of their stems and intersect the two measure-one sets above that stem. We call the resulting topology on q_* the *relative Prikry topology*. Its definition uses the ground forcing; it is not the product topology on enumerations of q_* .

Theorem 6.1 (Local constancy). Let $p \in W$. Every $T \subseteq q_*$ in OD_{p, q_*} is clopen in the relative Prikry topology. More generally, if $E \in W$ and $h : q_* \rightarrow E$ is OD_{p, q_*} , then h is locally constant when E has the discrete topology.

Proof. Fix $b \in q_*$. By Lemma 2.1, $W[b] = V$ and b is generic. Suppose first that $b \in T$. In the forcing language, use the formula that defines T from $p, [\dot{c}]$ and the fixed ordinal parameters. The truth lemma supplies a condition $P \in G_b$ forcing that its generic sequence belongs to the uniquely defined set. For any $d \in \mathcal{N}(P)$, we have $d \in q_*$ and $W[d] = V$. The formula therefore defines the original set T , and $P \in G_d$ forces $d \in T$. Thus $\mathcal{N}(P) \subseteq T$.

If $b \notin T$, apply the identical argument to $q_* \setminus T$. This proves openness of both T and its complement.

For the last assertion, $e = h(b)$ belongs to the ground-model set E , so $e \in W$. Apply the truth lemma to the statement that the uniquely defined function takes the value $\check{e}$ at the generic sequence. The same argument gives a neighborhood $\mathcal{N}(P)$ of b on which h is constantly e . No assumption that E is finite or small is needed. $\square$

An ordinal-valued definable map on q_* has the same property. One can apply the last statement with a ground ordinal larger than its range, since the ground and the extension have the same ordinals.

6.2 Every nonempty neighborhood has every phase

For $a, b \in q_*$ define the integer phase

$$\text{ind}_a(b) = |b \setminus a| - |a \setminus b| \in \mathbb{Z}. \quad (6.1)$$

Adding one new point to b raises its phase by one; deleting one point lowers it by one. These identities hold even if the edited point belongs to the reference set a . They follow either by separating the two finite symmetric differences or by comparing the lengths of prefixes preceding a common tail. Also

$$\text{ind}_a(b) + \text{ind}_b(d) = \text{ind}_a(d) \quad (a, b, d \in q_*).$$

Lemma 6.2 (Phase-coordinate geometry of a basic neighborhood). Suppose $\mathcal{N}(s, A)$ is nonempty, put $\ell = |s|$, and fix $a \in q_*$. For every $r \in \mathbb{Z}$ and every $n \geq \ell$,

$$\{b(n) : b \in \mathcal{N}(s, A), \text{ind}_a(b) = r\}$$

contains a tail of A . In particular its cardinality is κ . Moreover $|q_*| = \kappa$.

Proof. Fix $n \geq \ell$. Extend the finite sequence s by $n - \ell$ entries from A , obtaining a sequence s' of length n . When $n = 0$, use the empty sequence and any harmless lower bound in place of its last point. For each $\xi \in A$ above all entries of s' , let $v = s' \frown \langle \xi \rangle$, a sequence of length $n + 1$.

Choose a sufficiently late tail of a whose points all lie in A and all exceed ξ ; this is possible because a is Prikry generic. Append this tail to v , obtaining $b_0 \in \mathcal{N}(s, A) \cap q_*$ with $b_0(n) = \xi$. Let $r_0 = \text{ind}_a(b_0)$.

If $r \geq r_0$, insert $r - r_0$ fresh points of $A \setminus b_0$ above ξ . There are enough: A has cardinality κ , whereas b_0 is countable, and bounded initial segments have size less than κ . If $r < r_0$, delete $r_0 - r$ points of the infinite tail of b_0 above ξ . In either case the resulting b still has initial segment v , still realizes (s, A) , and differs only finitely from a . The phase identities give $\text{ind}_a(b) = r$, while $b(n) = \xi$ is unchanged.

This realizes every ξ in the indicated tail of A . Ground-model measure-one sets have ground size κ , and cardinal preservation keeps that size in V . Thus the projection has size κ .

Finally, every representative of q_* is a finite symmetric change of a , and every such change is still a cofinal set of order type ω . The bijection $F \mapsto a \Delta F$ from $[\kappa]^{<\omega}$ to q_* shows that $|q_*| = \kappa$. $\square$

Theorem 6.3 (Full phase and coordinate saturation). Let $T \subseteq D_\kappa$ be OD_{p, q_*} for $p \in W$, and suppose $T \cap q_* \neq \emptyset$. For every $a \in q_*$ and every $r \in \mathbb{Z}$,

$$|\{b \in T \cap q_* : \text{ind}_a(b) = r\}| = \kappa.$$

There is $n_0 < \omega$ such that, simultaneously for every integer r and every $n \geq n_0$,

$$|\{b(n) : b \in T \cap q_*, \text{ind}_a(b) = r\}| = \kappa.$$

The integer n_0 can be chosen independently of the reference $a \in q_*$.

Proof. By Theorem 6.1, the nonempty trace contains a nonempty basic neighborhood $\mathcal{N}(s, A)$. Put $n_0 = |s|$. Lemma 6.2 gives all of the lower bounds, for any chosen a and any integer phase. The cardinality $|q_*| = \kappa$ and the fact that coordinates lie below κ give the upper bounds. The same neighborhood works for every a . $\square$

Notice that cofinality alone would not give the asserted cardinality of a projection: $\text{cf}^V(\kappa) = \omega$. The proof instead exhibits a tail of a ground-model measure-one set in each projection. This is why the conclusion is full size κ , not merely unboundedness.

7 Rigidity and the one-measurable upper bound

For completeness, we recover the rigidity of q_* by the cone argument already used in the archive. This also verifies that the model used for the new finite-symmetry conclusions is the same kind of model as its one-measurable rigidity construction.

7.1 Tail-class-preserving cone isomorphisms

Lemma 7.1 (No new definable sets of ordinals from the tail class). If $p \in W$ and x is a set of ordinals in OD_{p,q_*}^V , then $x \in W$.

Proof. Given conditions (s, A) and (t, B) , choose $C \in U$ with $C \subseteq A \cap B$ and all its points above both stems. The cones below (s, C) and (t, C) are isomorphic by replacing the stem s with t and leaving every subsequently appended finite sequence and measure-one tail unchanged. Thus

$$(s \frown u, D) \mapsto (t \frown u, D).$$

The generic sequences in corresponding extensions differ by the finite stems s, t . Consequently the cone isomorphism preserves all check names, including p and ordinals, and preserves the name for the *tail class* of the generic sequence. This last assertion also follows directly by interpreting the two names in corresponding generic extensions: they produce the same universe and the same equivalence class.

It follows that the top condition decides every sentence of the form $\psi(\check{p}, [\dot{c}], \check{\alpha})$. Otherwise two conditions would force opposite values. Pass to the isomorphic sub-cones just described. The isomorphism transfers the first forced value to the second cone, contradicting its opposite forced value.

Choose an ordinal θ with $x \subseteq \theta$, and fix a formula defining x uniquely from $p, q_*, \check{\alpha}$. For each $\beta < \theta$, apply the preceding conclusion to the assertion that the uniquely defined object exists and contains β . In W , Separation using the forcing relation forms the set of $\beta < \theta$ for which the top condition forces that assertion. In the actual extension this set is precisely x , by uniqueness and the forcing theorem. Hence $x \in W$. $\square$

7.2 Short cofinal families cannot be definable

Let

$$C_\kappa = \{a \subseteq \kappa : \sup a = \kappa, |a|^V < \kappa\}.$$

Recall that rigidity of q_* means that every nonempty $\text{OD}_{V_\kappa, q_*}$ subfamily of C_κ has size at least κ .

Proposition 7.2 (Rigidity, with ground parameters). For every $p \in W$, every nonempty $\mathcal{A} \subseteq C_\kappa$ in OD_{p, q_*} has $|\mathcal{A}|^V \geq \kappa$. In particular q_* is rigid.

Proof. Suppose instead that $0 < |\mathcal{A}|^V < \kappa$. Let η be the least order type of a member of $\mathcal{A}$, and put

$$\mathcal{A}_0 = \{a \in \mathcal{A} : \text{ot}(a) = \eta\}, \quad u = \bigcup \mathcal{A}_0.$$

Every member of C_κ has order type below the initial ordinal κ , so $\eta < \kappa$. The set u is unbounded in κ , is OD_{p, q_*} , and satisfies

$$|u|^V \leq |\mathcal{A}|^V \cdot |\eta|^V < \kappa.$$

The last inequality holds for the product of two cardinals below an infinite cardinal, even though κ is singular in V .

By Lemma 7.1, $u \in W$. Unboundedness of a given set of ordinals in κ is absolute between these two transitive models. Since κ is regular in W , an unbounded ground set has ground size κ . A ground bijection witnessing that size remains a bijection in V , contradicting the displayed inequality.

Every finite tuple of parameters from V_κ^V belongs to W , so the last assertion follows. $\square$

Corollary 7.3 (Regularity in the relativized inner model). The cardinal κ is regular in $N_{q_*} = \text{HOD}_{V_\kappa \cup \{q_*\}}$, although $\text{cf}^V(\kappa) = \omega$.

Proof. If N_{q_*} had a cofinal map $f : \delta \rightarrow \kappa$ for some $\delta < \kappa$, then f and its range would be ordinal-definable from q_* and finitely many individual parameters in V_κ . The range would be an unbounded set of size at most $|\delta| < \kappa$. Lemma 7.1 places it in W , giving the same contradiction to ground regularity and cardinal preservation as above. This proof does not assume Choice in N_{q_*} . $\square$

7.3 What consistency conclusion has actually been proved

Corollary 7.4 (One-measurable construction). If ZFC with a measurable cardinal is consistent, then so is ZFC with a strong limit cardinal κ of cofinality ω and a rigid $q_* \in Q_\kappa$ satisfying all conclusions of Theorem 1.1.

Proof. A measurable cardinal carries a normal measure; force with its normal Prikry forcing. Input $\mathcal{A}$ gives the strong-limit, cofinality, and rank-agreement assertions. Proposition 7.2 gives rigidity. Theorems 4.3, 5.1, and 6.3 give the finite-symmetry and saturation conclusions. The ultrafilter assertion is established in Section 8. These proofs are forcing arguments over ZFC; the transitive-model notation used for exposition imposes no additional transitive-model consistency assumption. $\square$

The supplied synthesis gives an exact measurable calibration for a weaker package including rigidity, using quoted core-model covering theorems. The construction here extends its *upper-bound* side to the finite selector and finite-label obstructions. No new lower-bound proof is offered here, and none is needed for the new same-universe recoding theorem. In particular, no part of the argument asserts that a model with an ultraextending cardinal is obtainable from one measurable cardinal.

8 A sharp countable threshold for local ultrafilters

An ultrafilter in this section is an ultrafilter on the full Boolean algebra $\mathcal{P}^V(q)$ of subsets of a tail class q . This is not a normal measure on κ inside an inner model. Keeping these two domains separate is essential.

8.1 No finite definable family at the Prikry class

Theorem 8.1 (Finite local ultrafilter obstruction). For every $p \in W$, there is no nonempty finite OD_{p, q_*} family of ultrafilters on $\mathcal{P}^V(q_*)$.

Proof. Suppose $\mathcal{U}$ is such a family, with $M = |\mathcal{U}|$. Fix an integer $N > M$. For a representative $a \in q_*$, partition q_* into the phase residue classes

$$P_i(a) = \{b \in q_* : \text{ind}_a(b) \equiv i \pmod{N}\}, \quad i \in \mathbb{Z}/N\mathbb{Z}.$$

Every ultrafilter contains exactly one cell of a finite partition. Thus

$$S(a) = \{i \in \mathbb{Z}/N\mathbb{Z} : P_i(a) \in \mathcal{U} \text{ for some } \mathcal{U} \in \mathcal{U}\}$$

is nonempty and has size at most M .

Apply Lemma 2.3 to the finite observation $S(a)$. Let $P \in G_c$ decide its value to be $S(c)$, and delete one point of c after the stem of P , obtaining d . The recoding lemma leaves V, q_* , and the definable family $\mathcal{U}$ unchanged, and yields $S(d) = S(c)$. But $\text{ind}_c(d) = -1$, so the phase cocycle gives

$$\text{ind}_d(b) = \text{ind}_c(b) + 1 \quad (b \in q_*).$$

Therefore $P_i(d) = P_{i-1}(c)$ and $S(d) = S(c) + 1$ modulo N . A nonempty subset of $\mathbb{Z}/N\mathbb{Z}$ invariant under addition of 1 is the whole set. Hence $|S(c)| = N > M$, a contradiction. $\square$

Corollary 8.2 (Finite global-kernel obstruction). There is no nonempty finite OD_{p,q_*} family of functions K on Q_κ such that $K(q)$ is an ultrafilter on $\mathcal{P}^V(q)$ for every $q \in Q_\kappa$.

Proof. Evaluate every member of the family at q_* . The resulting image is a nonempty finite OD_{p,q_*} family of ultrafilters on q_* , contrary to Theorem 8.1. $\square$

8.2 A countable local family in ordinary ZFC

The positive construction does not require Prikry forcing or a measurable cardinal. It uses a free ultrafilter on the integers and all of its shifts, so that a finite change of representative only reindexes the countable family.

Fix a free ultrafilter D on $\mathbb{Z}$ containing ω . Such an ultrafilter exists in ZFC, by transporting a free ultrafilter on ω to $\mathbb{Z}$. For $k \in \mathbb{Z}$, let

$$D_k = (\text{sh}_k)_* D, \quad \text{sh}_k(n) = n + k,$$

where the pushforward of an ultrafilter under $f : X \rightarrow Y$ is

$$f_* D = \{A \subseteq Y : f^{-1}[A] \in D\}.$$

Every D_k is free and contains ω , since translates of ω differ from it by only finitely many integers.

For $a \in D_\lambda$, define a map $t_a : \mathbb{Z} \rightarrow [a]$ by

$$t_a(n) = \begin{cases} \{a(j) : j \geq n\}, & n \geq 0, \\ a, & n < 0. \end{cases}$$

Its values at negative indices are irrelevant to all the D_k . Set

$$\mathcal{U}_D([a]) = \{(t_a)_* D_k : k \in \mathbb{Z}\}. \quad (8.1)$$

Theorem 8.3 (Countable tail-supported ultrafilter correspondence). For every uncountable cardinal λ of cofinality ω , formula (8.1) defines a function on Q_λ , independent of the chosen representative a . For every $q \in Q_\lambda$, the set $\mathcal{U}_D(q)$ is countably infinite and consists of free ultrafilters on the full $\mathcal{P}(q)$. Every one of these ultrafilters contains the tail ray

$$R_b = \{t_b(n) : n < \omega\} \quad \text{for every } b \in q.$$

The function is definable from D and λ , and its value at q is $\text{OD}_{D,q}$. The parameter D belongs to V_λ .

Proof. Existence and full domain. For fixed a, k , the pushforward $(t_a)_*D_k$ is an ultrafilter on the entire $\mathcal{P}([a])$: for every subset $B \subseteq [a]$, its inverse image is a subset of $\mathbb{Z}$, and D_k decides that inverse image or its complement. This is not an ultrafilter merely on a specified subalgebra. The map t_a is injective on ω , so the inverse image of a singleton meets ω in at most one point. Because D_k is free and contains ω , no singleton belongs to the pushforward.

Independence of the representative. If $b =^* a$, their increasing enumerations have synchronized tails. More explicitly, choose $I, J < \omega$ such that

$$a(I + n) = b(J + n) \quad (n < \omega).$$

With $e = I - J$, this gives $t_b(n) = t_a(n + e)$ for all sufficiently large n . Free ultrafilters containing ω disregard the exceptional indices. Therefore

$$(t_b)_*D_k = (t_a)_*D_{k+e} \quad (k \in \mathbb{Z}).$$

Since $k \mapsto k + e$ permutes $\mathbb{Z}$, the sets of pushforwards in (8.1) are equal. This proves independence and also shows that the value is definable from D, q , without an individual representative as a parameter.

Distinctness of all shifts. No free ultrafilter on $\mathbb{Z}$ is fixed by a nonzero integer translation. To see this, suppose it were fixed by translation by $h > 0$. Let

$$E_h = \{n \in \mathbb{Z} : n \bmod 2h \in \{0, 1, \dots, h-1\}\}.$$

Translation by h exchanges E_h and its complement. Invariance would make an ultrafilter contain both or neither, which is impossible. A negative period implies the corresponding positive period. It follows that $D_k \neq D_l$ whenever $k \neq l$.

Pushforward by t_a preserves this distinction. Indeed, all D_k concentrate on ω , where t_a is injective. For $B \subseteq \omega$, the inverse image of $t_a[B]$ agrees with B on ω . Consequently one can recover every membership decision of D_k on ω from its pushforward, and decisions on ω determine an ultrafilter containing ω . Thus the pushforwards are pairwise distinct. Formula (8.1) gives exactly $\aleph_0$ of them.

Tail support. Each $(t_a)_*D_k$ contains R_a , since $t_a^{-1}[R_a]$ contains ω . If $b =^* a$, the synchronized-tail identity shows that R_a and R_b differ by finitely many elements of q . A free ultrafilter containing R_a therefore contains R_b as well.

Rank and definability. An ultrafilter on $\mathbb{Z}$ is a set of subsets of a countable low-rank set. Its rank, under the usual coding of integers, is below $\omega + \omega$. Hence $D \in V_\lambda$. Independence of representatives makes (8.1) a functional definition on the set Q_λ ; Replacement supplies its graph. All the claimed definability statements now follow from the explicit formula. $\square$

Corollary 8.4 (Exact local threshold). In the Prikry setup, the least size of a nonempty $\text{OD}_{V_\kappa, q_*}$ family of ultrafilters on $\mathcal{P}(q_*)$ is $\aleph_0$. There is such a countable family all of whose ultrafilters are free and contain every representative's tail ray.

Proof. Theorem 8.1 rules out every positive finite size, since all permitted low-rank parameters belong to W . Choose D as in Theorem 8.3. It belongs to V_κ , and that theorem gives a countably infinite family with all the required properties. $\square$

8.3 Why this does not solve the global countable-kernel question

Theorem 8.3 gives one definable *correspondence*

$$q \mapsto \mathcal{U}_D(q), \quad |\mathcal{U}_D(q)| = \aleph_0.$$

It does not give a definable countable set of functions $\{K_n : n < \omega\}$ with $K_n(q)$ an ultrafilter on q simultaneously for all q . The displayed indexing in (8.1) uses a representative a . Replacing a changes that indexing by an integer shift which can depend on q . The set of values is canonical, but the index origin is not.

Ordinary Choice can select representatives and synchronize these indexings, but it does not make the resulting synchronization definable from D or other permitted parameters. Therefore the countably infinite global-family question in the supplied synthesis is not resolved by this construction. The local threshold in Corollary 8.4 is an exact theorem of a different, explicitly stated kind.

9 Proof audit and the remaining boundary

9.1 Dependencies and new content

The new forcing argument depends on normal Prikry preservation, the Mathias criterion, and the ordinary forcing theorem. It does not depend on an ultraexacting witness, a correctness-height argument, an iterated ultrapower, a core-model covering theorem, or the strongly compact sandwich results in the supplied reports.

The combinatorial amplification and the incidence-word sufficiency construction have antecedents in the archive. The new step is the finite-table recoding lemma in a normal Prikry extension, together with its applications to finite families of arbitrary finite equivariant rules. The local clopen theorem and its full-sized phase-coordinate corollary isolate a further consequence of the same mechanism. The countable ultrafilter correspondence supplies a sharp local positive counterpart to the finite obstruction.

This is new relative to the mathematical development in the supplied archive: it proves the previously unproved Prikry-model assertion asked there and strengthens it to allow both the generic quotient parameter and arbitrary fixed ground-model parameters. The literature check verified the classical forcing inputs and the large-cardinal comparison used for context. It did not establish bibliographic priority for every application or rule out equivalent formulations elsewhere.

9.2 Checks that prevent false strengthenings

Six distinctions maintained throughout the proofs

One class versus one representative. The parameter q_* is fixed under finite recoding; c is not available as a defining parameter. Allowing c would destroy the central symmetry.

A definable family versus definable members. The finite set of evaluation tables is decided as a set. Members of the original family need not be individually definable or fixed.

A Prikry model versus abstract rigidity. We prove the finite obstructions in a concrete forcing extension and also prove that its generic class is rigid. We do not deduce the recoding property from rigidity alone.

Elementwise versus common fixed points. Each $g \in \Gamma$ must fix some label, which may depend on g . The five-label S_3 example prevents replacing this with a common fixed label requirement.

Local ultrafilters versus inner-model measures. The ultrafilters in Section 8 decide all subsets of q_* in V . They are not normal measures on κ in a relativized HOD.

Countably many values versus countably many global functions. A countable-valued correspondence on Q_λ does not provide a definable simultaneous enumeration of its fibres.

9.3 Questions that are still genuinely open in this continuation

The next structural problem is whether the finite-table recoding conclusion follows from some intrinsic strengthening of rigidity, weaker than a full normal Prikry presentation. This would distinguish which part of the phenomenon is topological and which part is forced by regularity in N_q .

The unrestricted implication from rigidity to the finite-selector obstruction remains unproved here. So do the questions about ordinary exactingness, countably infinite definable families of global selectors, and countably infinite definable families of global ultrafilter-valued kernels at an ultraexacting cardinal. Neither the positive local ultrafilter theorem nor the one-measurable construction decides them.

The proved advance is therefore specific but substantial: the source archive's finite-symmetry boundary is no longer confined to the ultraexacting construction. It is already forced by finite recodings of an ordinary normal Prikry generic sequence.

A Finite verification and reproducibility

The archive's Lean file `Cardinals/Combinatorics/FiniteCycles.lean` separates finite permutation amplification from the large-cardinal interface. That separation guided Section 3. No new Lean theorem is claimed here, and the supplied Lean development was not recompiled for this report.

The accompanying Python script `finite_checks.py` uses only the standard library. It performs exact finite checks of three ingredients: it exhaustively computes cyclic orbits of small selector tables, checks the amplified permutation embedding for every permutation of at most six labels and family-size bounds at most five, and verifies the five-label S_3 fixed-point example. It also checks that finite-word stabilizers with faithful label separation are cyclic, for the small periodic examples described in its output.

The checks are deterministic. They neither simulate a measurable cardinal nor verify a forcing argument, and they are not a substitute for the English proofs. Their purpose is to detect an incorrect permutation convention, an insufficient amplification length, or an accidental common-fixed-point assumption.

Finite test	Exact value	Interpretation
Binary selectors on 2 points	2 tables; minimum cyclic orbit 2	The two choices can be exchanged; the unamplified test does not exclude a 2-member family
Binary selectors on 4 points	64 tables; minimum cyclic orbit 4	The 4-cycle amplification excludes invariant families of size at most 2
Binary selectors on 6 points	32768 tables; minimum cyclic orbit 2	An arbitrary large cycle is insufficient; its length must match the amplification construction
Three-to-one selectors on 3 points	3 tables; minimum cyclic orbit 3	A 3-cycle has no invariant singleton choice
Five-label S_3 target	Each of 6 permutations fixes a label; no common fixed label	The theorem uses the elementwise condition, not a global fixed point

The distribution of all orbit sizes and all other test counts are recorded in `finite_checks.txt`. The script's assertions fail with a nonzero exit status if a check disagrees with the expected identity. Its input sizes are deliberately bounded so that it runs quickly on an ordinary Python installation.

Build

The report uses the same `newpxtext/newpxmath` font setup, page geometry, theorem style, and Forest–Olive–Sage color definitions as the supplied synthesis. A standard TeX installation with those packages can rebuild the PDF by running

```
pdflatex -interaction=nonstopmode -halt-on-error Prikry_Finite_Symmetry.tex
pdflatex -interaction=nonstopmode -halt-on-error Prikry_Finite_Symmetry.tex
python3 finite_checks.py
```

An additional LaTeX pass may be used if a local engine requests it for updated cross-references. Font files are not redistributed.

References

- [1] *Large cardinals at the inconsistency frontier: a synthesis*. Supplied research archive Cardinals4.zip, docs/Large_Cardinals_Synthesis.tex, dated 18 September 2026. Especially Sections 9, 12.3, 12.5, 13.4–13.5, and the final question of Section 15. Consulted together with docs/Large_Cardinals_Unified_Report.tex and the accompanying Cardinals/ Lean sources. Internal research material; not presented here as an independently peer-reviewed source.
- [2] A. R. D. Mathias, *On sequences generic in the sense of Prikry*, Journal of the Australian Mathematical Society **15** (1973), 409–414. doi:10.1017/S1446788700028755.
- [3] Tom Benhamou, *Prikry forcing and tree Prikry forcing of various filters*, Archive for Mathematical Logic **58** (2019), 787–817. doi:10.1007/s00153-019-00660-3. Open version: arXiv:1801.04424, version 2, 25 September 2018. The theorem numbers in this report refer to that open version.
- [4] Juan P. Aguilera, Joan Bagaria, Gabriel Goldberg, and Philipp Lücke, *Large cardinals beyond HOD*, arXiv:2509.10254, version 1, 12 September 2025. Used for the contextual comparison with I_0 , not as an input to the new finite-tail recoding proofs.
- [5] Juan P. Aguilera, Joan Bagaria, and Philipp Lücke, *Large cardinals, structural reflection, and the HOD Conjecture*, arXiv:2411.11568, version 4, 16 September 2025. Background for the exacting and ultraexacting framework continued by the supplied archive.